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shaping futures

CS201 – Data Structures II

Lecture 14

Dr. Muhammad Mobeen Movania

(Content has been taken from the lecture notes of
Abdullah Zafar and Dr. Syeda Saleha Raza)



Outlines

- Motivation
- Parametric and zone indexes
- Compressed zone indexes
- Weighted zone scoring
- Term frequency (TF)
- Inverse document frequency (IDF)

Motivation

- We have dealt with indexes that support Boolean queries
 - A document either matches or does not match a query
- It is essential for a search engine to rank-order the documents matching a query.
- Search engine computes, for each matching document, a score with respect to the query at hand.
- Scoring may be obtained through:
 - Parametric and zone indexes: that is we can index and retrieve documents using metadata for e.g. language in which a document is written, author of the document etc.
 - Weighting the importance of a term in the document based on its occurrences.
 - Viewing each document as a vector of weights.

Parametric and zone indexes

- Metadata contains fields like author, title, year etc. which are finite length strings.
- Query processing then consists as usual of postings intersections, except that we may merge postings from standard inverted as well as ***parametric indexes***.
 - There is one parametric index for each field (for e.g. date of creation); it allows us to select only the documents matching a date specified in the query
- *Zones* are similar to fields, except the contents of a zone can be arbitrary free text
 - For e.g. author, title
 - A separate inverted index is generated for each zone

Example: Zones and Fields

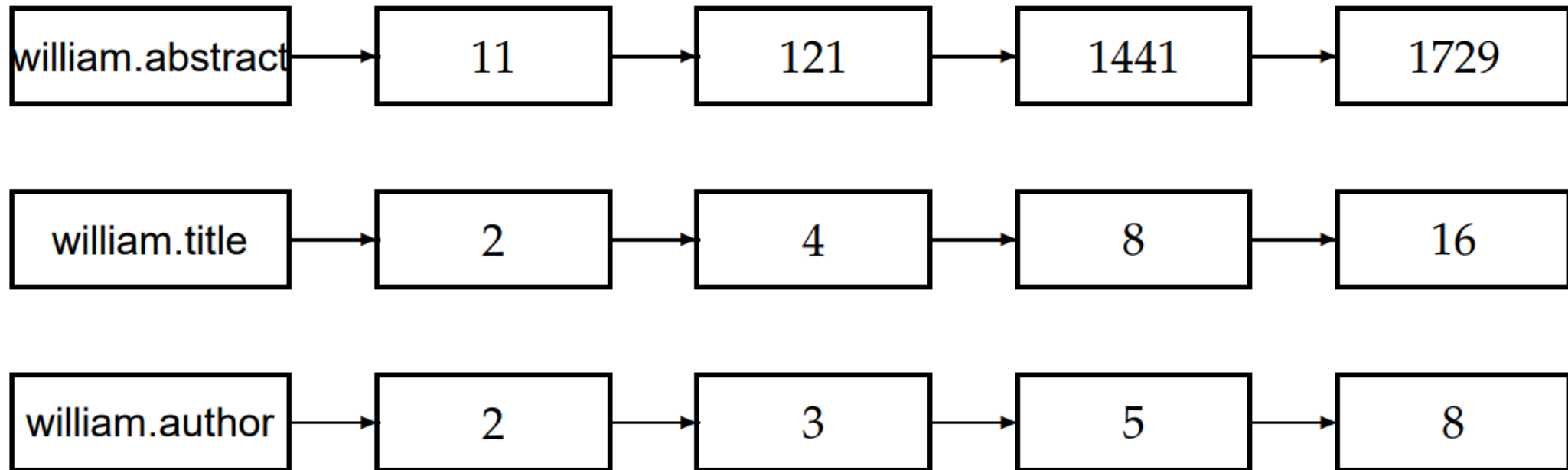
Bibliographic Search

Search category	Value
<u>Author</u>	Example: Widom, J or Garcia-Molina
<u>Title</u>	Also a part of the title possible
<u>Date of publication</u>	Example: 1997 or <1997 or >1997 limits the search to the documents appeared in, before and after 1997 respectively
Language	Language the document was written in English v
Project	ANY v
Type	ANY v
Subject group	ANY v
Sorted by	Date of publication v

Zones {

Fields {

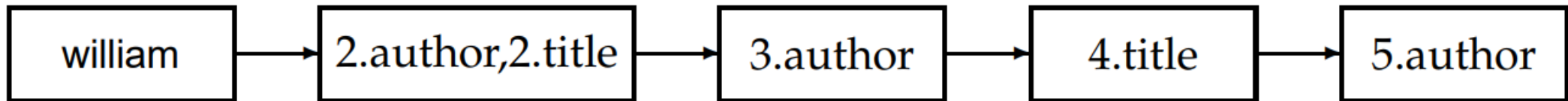
Example: Separate inverted index for each zone



► **Figure 6.2** Basic zone index ; zones are encoded as extensions of dictionary entries.

Compressed inverted index for each zone

- We can reduce the size of the dictionary by encoding the zone in which a term occurs in the postings.
- This compressed representation also helps in efficient computation of scores using a technique called ***weighted zone scoring***.



► **Figure 6.3** Zone index in which the zone is encoded in the postings rather than the dictionary.

Weighted zone scoring

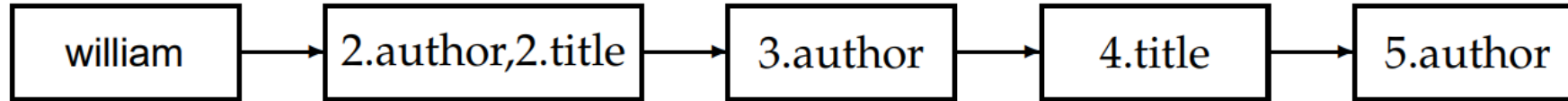
- Weighted zone scoring (aka Ranked Boolean retrieval) assigns to the pair (q, d) for a Boolean query (q) and a document (d), a score in the interval [0, 1], by computing a linear combination of zone scores, where each zone of the document contributes a Boolean value.
- Given mathematically as follows:

$$\sum_{i=1}^l g_i s_i$$

- Where, l is the total number of zones,
- s_i is the Boolean score denoting a match (or absence thereof) between q and the i^{th} zone
- g_i are the weights assigned to each zone, such that $\sum_{i=1}^l g_i = 1$, the higher the value, the more the significance of that zone

Exercise

- Suppose we have the following compressed inverted index.



- Zone weights are given as follows:
 - g_1 (title) = 0.4
 - g_2 (author) = 0.1
 - g_3 (date) = 0.1
 - g_4 (body) = 0.4
- The search term “William” is assigned the following scores.

Document	1	2	3	4	5
Score	0	0.5	0.1	?	0.1

- Calculate the score for document 4?
- Which document appears at the top for search “William”?

Issues with weighted zone scoring

- Scoring is hinged on whether or not a query term is present in a zone within a document.
- Depending on the zone in which the query term exist, the corresponding weight is assigned to the query term
- What if we score a term based on its frequency in the document or zone?

Term Frequency (TF)

- **Key Idea:** A document or zone that mentions a query term more often has more to do with that query and therefore should receive a higher score.
- In term frequency (TF), we assign a weight to each word in a document based on the frequency of the word in the document.
 - Given as: $tf_{t,d}$ for “term frequency of term t in document d ”.
- In the bag of words model, the exact ordering of terms is ignored, only the frequency of each term is stored.
- More sophisticated methods have more discriminating power.

Example

- You are given the (document, $tf_{t,d}$) data as follows:

the	state	of	exception	is	normal
(1,9)	(3,3)	(1,5)	(3,3)	(1,10)	(4,3)
(3,2)	(4,1)	(2,7)	(4,1)	(2,3)	
(4,10)		(3,5)		(4,5)	
		(4,10)			

- Given query “state AND exception” generates the following table

Document	1	2	3	4
Score	0	0	3+3=6	1+1=2

- So in terms of score, document 3 will be ranked on top

Term Frequency: Issues

- All terms are considered equally important when it comes to assessing relevancy on a query.
- Certain terms have little or no discriminating power in determining relevance. For e.g: a document from the auto industry will have the word auto used frequently.
- How to cater for above issues?
 - We divide the term frequency by the **collection frequency** (which is total number of occurrence of a term in the collection)
 - A more common metric is **document frequency (df)** which is the number of documents in the collection that contain the term t.

Word	cf	df
try	10422	8760
insurance	10440	3997

► **Figure 6.7** Collection frequency (cf) and document frequency (df) behave differently, as in this example from the Reuters collection.

Inverse Document Frequency (IDF)

- How is the document frequency df of a term used to scale its weight?
- We define the ***inverse document frequency (idf)*** of a **term t** as follows:

$$idf_t = \log\left(\frac{N}{df_t}\right)$$

- Where,
 - N is the total number of documents in the collection
 - df_t is the document frequency of term t
- Idf of a rare term is high

Tf-idf Weighting

- A composite weight is produced using the definitions of term frequency and inverse document frequency, for each term in each document

$$tf_idf_{t,d} = tf_{t,d} \times idf_t$$

- $tf_idf_{t,d}$ assigns to term t a weight in document d that is
 - highest when t occurs many times within a small number of documents (thus lending high discriminating power to those documents)
 - lower when the term occurs fewer times in a document, or occurs in many documents (thus offering a less pronounced relevance signal)
 - lowest when the term occurs in virtually all documents

- Overlap score measure ($Score(q,d)$)

$$Score(q, d) = \sum_{t \in q} tf_idf_{t,d}$$

Example

- Document frequencies (df_t) of some terms are given below:

Term	the	state	of	exception	is	normal
df_t	3	2	4	2	3	1

- $N = 6$ (6 documents in collection)

Term	the	state	of	exception	is	normal
N/df_t	$6/3=2$	$6/2=3$	$6/4=1.5$	$6/2=3$	$6/3=2$	$6/1=6$

- Take $\log_{10} \left(\frac{N}{df_t} \right)$ to get idf_t

Term	the	state	of	exception	is	normal
N/df_t	0.301	0.477	0.176	0.477	0.301	0.778



Further Reading

- **Introduction to Information Retrieval** by Christopher D. Manning, Prabhakar Raghavan & Hinrich Schütze, Cambridge University Press, 2008. Available online: [Introduction to Information Retrieval \(stanford.edu\)](#)
 - Chapter 6
- Abdullah Zafar's lecture notes on inverted index. Available online: [Week 13 - Inverted Index - Google Docs](#)



Next Lecture

- Lecture 15